Day21 - 11th May

S3 Storage Classes

📊 S3 Storage Classes Overview

Storage Class	Use Case	Retrieval Time	Approx Cost
S3 Standard	Frequently accessed data	Milliseconds	Highest
S3 Intelligent-Tiering	Unknown access patterns	Milliseconds	Medium
S3 Standard-IA	Infrequent access	Milliseconds	Lower
S3 One Zone-IA	Infrequent + single AZ	Milliseconds	Cheaper
Glacier Instant Retrieval	Rare access	Milliseconds	Very cheap
Glacier Flexible Retrieval	Archive	Minutes/Hours	Very low
Glacier Deep Archive	Long-term backup	Hours	Cheapest

S3 one Zone IA - not recommended

🔥 1️⃣ S3 Standard

Use Case

- Websites
- Apps
- Frequently accessed files

Features

✅ Multi-AZ
✅ Low latency
✅ High throughput

2. Infrequent
3. Glacier instant retrieval
4. Glacier flexible retrieval
5. Deep glacier

S3 standard frequent

1. Default storage class in standard
2. We can upload any size of the data no restrictions
3. No waiting period required
4. Anytime we can access data no download charges

Infrequent

1. Not frequently access data
2. minimum 30 days waiting period required to get benefits

ex: if we upload a file today infrequent class immediately 30 days billing will be considered if delete file tomorrow still pricing will be applied for 30 days. if I keep my file 40 days you'll

Glacier

minimum waiting period is 90 days

Deep glacier

minimum waiting period is 180 days

🔥 3️⃣ S3 Standard-IA (Infrequent Access)

Use Case

- Backups
- DR files
- Monthly access

Features

✅ Multi-AZ
✅ Lower storage cost
❌ Retrieval charges apply

Approx Cost

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~$0.0125 per GB/month
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📊 S3 Storage Class Discount Comparison

Storage Class	Approx Cost/GB	Approx Savings vs Standard
S3 Standard	$0.023	0%
Intelligent-Tiering	~$0.023	0–10%
Standard-IA	~$0.0125	~45% cheaper
One Zone-IA	~$0.01	~57% cheaper
Glacier Instant Retrieval	~$0.004	~82% cheaper
Glacier Flexible Retrieval	~$0.0036	~84% cheaper
Glacier Deep Archive	~$0.00099	~96% cheaper

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Amazon S3 > Buckets > cloudfronts3static > Edit storage class

Amazon S3 offers a range of storage classes designed for different use cases. Learn more or see Amazon S3 pricing

Storage class	Designed for	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
Standard	Frequently accessed data (more than once a month) with milliseconds access	≥ 3	-	-	-	-
Intelligent-Tiering	Data with changing or unknown access patterns	≥ 3	-	-	Per-object fees apply for objects >= 128 KB	-
Standard-IA	Infrequently accessed data (once a month) with milliseconds access	≥ 3	30 days	128 KB	-	Per-GB fees apply
One Zone-IA	Recreatable, infrequently accessed data (once a month) with milliseconds access	1	30 days	128 KB	-	Per-GB fees apply
Glacier Instant Retrieval	Long-lived archive data accessed once a quarter with instant retrieval in milliseconds	≥ 3	90 days	128 KB	-	Per-GB fees apply
Glacier Flexible Retrieval (formerly Glacier)	Long-lived archive data accessed once a year with retrieval of minutes to hours	≥ 3	90 days	-	-	Per-GB fees apply
Glacier Deep Archive	Long-lived archive data accessed less than once a year with retrieval of hours	≥ 3	180 days	-	-	Per-GB fees apply
Reduced redundancy	Noncritical, frequently accessed data with milliseconds access (not recommended as S3 Standard is more cost effective)	≥ 3	-	-	-	Per-GB fees apply

🚀 Difference from Other Glacier Types

Storage Class	Retrieval Time
Glacier Instant Retrieval	Milliseconds
Glacier Flexible Retrieval	Minutes/Hours
Glacier Deep Archive	Hours

S3 lifecycle

S3 lifecycle rule we can use to automate the process
like move file diff storage classes
or delete files current version or non current version